

June 22, 2026

FINATRAX Research Group
Interdisciplinary Centre for Security, Reliability and Trust (SnT)
University of Luxembourg

Dear Members of the Selection Committee,

I am writing to formally submit my application for the Doctoral Researcher position focusing on the Application of Large Language Models and Knowledge Graphs for task automation. As an AI & Data Science Engineer, my research is entirely focused on bridging the gap between highly parameterized "black-box" neural networks and strict, structured rule systems.

During my engineering theses, I focused heavily on ensuring that AI models do not hallucinate when deployed in critical environments. For my Master's thesis on the "Cardiovascular Medical Digital Twin," I engineered Medical_MambaTS, leveraging linear-time state-space models. More importantly, I developed PI-DeepONet, a Physics-Informed Operator Network that explicitly maps rigorous mathematical constraints (differential equations) into the neural architecture. This experience of embedding structural, deterministic bounds into neural networks perfectly aligns with your objective of using Knowledge Graphs to ground LLMs in reality, ensuring that automated tasks are executed safely and predictably.

I am highly motivated to transition my expertise in structured neural networks to the domain of Information Systems Engineering. The challenge of deploying hybrid LLMs that can securely interact with business automation tools—constrained by socio-technical requirements and Knowledge Graph architectures—is exactly the interdisciplinary challenge I wish to pursue at FINATRAX.

I possess strong programming skills in Python, deep experience with cloud-ready neural architectures, and a rigorous theoretical foundation. Please find attached my CV and academic transcripts for your review.

I look forward to the possibility of discussing how my background aligns with your research goals.

Sincerely,
Ismail Aissaoui
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